Group II, Row IV: the $d_\perp$ term and the plus prescription

Your expressions are in black, new ones in blue. The term is the third line of Eq. (fin) of `RowIV_FF_userform.pdf`.

1 The term

$$\left.\frac{d^3N}{d^3k}\right|_{d_\perp} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+\!+\Lambda}} d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} \times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big] \times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{1.1}$$

- No pole multiplies this term, so $d_\perp = 2-\epsilon$ can be set to 2 here (the difference is $O(\epsilon)$). With $X \equiv \frac{x-y}{(x-y)^2}$, $X' \equiv \frac{x'-y'}{(x'-y')^2}$ and $\hat\Delta \equiv \frac{y'-y}{|y'-y|}$, the bracket is

$$X\cdot X' - 2(X\cdot\hat\Delta)(X'\cdot\hat\Delta) = X^iX'^j\big(\delta^{ij}-2\hat\Delta^i\hat\Delta^j\big). \tag{1.2}$$

  The tensor $\delta^{ij}-2\hat\Delta^i\hat\Delta^j$ is traceless. This is what decides the answer below.

- Write

$$F(\xi) \equiv \int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big] \times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{1.3}$$

  so that Eq. (1.1) is $\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)} d\xi\,\frac{1-\xi}{\xi}F(\xi)$.

- With $\xi = \frac{k^+}{p^+\!+k^+}$ (so $d\xi\,\frac{1-\xi}{\xi} \to dp^+\frac{p^+}{(p^+\!+k^+)^2}$):

$$\left.\frac{d^3N}{d^3k}\right|_{d_\perp} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\Lambda}^{\vee-k^+} dp^+\,\frac{p^+}{(p^+\!+k^+)^2}\,F\Big(\frac{k^+}{p^+\!+k^+}\Big). \tag{1.4}$$

2 Answer

Yes, you may write a plus prescription, but it subtracts zero at both ends. The term has no $\log\frac{\vee}{\Lambda}$, no $\log\frac{\vee}{k^+}$ and no $\log\frac{k^+}{\Lambda}$. It is finite and goes entirely into the finite part.

- $\xi\to 1$ **($p^+\to 0$, the Λ end).** The integrand is $\propto(1-\xi)$: in Eq. (1.4) it is $\frac{p^+}{k^{+2}}F(1)\to 0$.
  - There is no $\frac{1}{1-\xi}$ (no $\frac{1}{p^+}$), so there is nothing for $\big[\frac{\xi}{1-\xi}\big]_+$ or for your $\big[\frac{k^+}{p^+}\big]_+$ to regulate.
  - In your definition, $\frac{p^+}{P^2}F = \frac{k^+}{p^+}f(p^+)$ with $f(p^+) = \frac{p^{+2}}{k^+P^2}F$ and $f(0)=0$. Hence $\big[\frac{k^+}{p^+}\big]_+ f = \frac{k^+}{p^+}f$, and the term $f(0)\log\frac{k^+}{\Lambda}$ it would generate is zero.
- $\xi\to 0$ **($p^+\to\vee$, the $\vee$ end).** In Eq. (1.4) the weight is $\frac{1}{p^+}$, so there would be a $\log\vee$ if $F(0)\neq 0$. But (Sec. 3)

$$F(\xi) = \frac{8\pi\,\xi^2}{\mathbf{k}^2}\mathcal{C} + O(\xi^3),$$
$$\mathcal{C} \equiv \int_{x,x',y}\frac{(x-y)\cdot\hat{\mathbf{k}}\,(x'-y)\cdot\hat{\mathbf{k}} - \frac{1}{2}(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}\big[U^{e'c}(y)-U^{e'c}(x'\quad)\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{2.1}$$

  - The integrand $\frac{1-\xi}{\xi}F(\xi)$ is $O(\xi)$, i.e. $O(1/p^{+3})$ in Eq. (1.4).
  - A plus prescription at $\xi=0$, $\big[\frac{1}{\xi}\big]_+$ acting on $f(\xi)=(1-\xi)F(\xi)$, subtracts $f(0)=0$. The $\delta(\xi)$ term it would generate, $f(0)\log\frac{\vee}{k^+}$, is zero.

Result. Both limits can be moved to 0 and 1, with power corrections only:

$$\boxed{\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{d_\perp} &= \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} \\ &\times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big] \\ &\times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x) \\ &-\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\Big[\frac{\Lambda^2}{2(k^++\Lambda)^2}\,F(1)+\frac{4\pi}{\mathbf{k}^2}\frac{k^{+2}}{\vee^2}\,\mathcal{C}\Big] \\ &+O\Big(\frac{\Lambda^3}{k^{+3}},\frac{k^{+3}}{\vee^3}\Big).\end{aligned}} \tag{2.2}$$

- $F(1)=\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\big[\dots\big]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x)$ is F at $\xi=1$.
- With $\Lambda\to 0$ and $\vee\to\infty$ only the integral from 0 to 1 remains. It depends on neither Λ nor $\vee$, so it is part of the finite (scale-independent) cross section.
- It enters neither JIMWLK, nor BFKL, nor DGLAP (it has no pole).

3 Why $F(\xi)=O(\xi^2)$

With $\Delta\equiv y'-y$,

$$\begin{aligned}F(\xi) &= \int d^2\Delta\, e^{-ik\cdot\Delta/\xi}\big(\delta^{ij}-2\hat\Delta^i\hat\Delta^j\big)\,\mathcal{G}^{ij}(\Delta), \\ \mathcal{G}^{ij}(\Delta) &\equiv \int_{x,x',y}\frac{(x-y)^i}{(x-y)^2}\frac{(x'-y-\Delta)^j}{(x'-y-\Delta)^2}\,\mathcal{O}\big|_{y'=y+\Delta},\end{aligned} \tag{3.1}$$

with $\mathcal{O}$ the colour factor of Eq. (1.1). The parent momentum is $\mathbf{q}=\mathbf{k}/\xi$, so $\xi\to 0$ is the Fourier transform at large $|\mathbf{q}|$. It is governed by $\Delta\to 0$, where $\mathcal{G}^{ij}(\Delta)=\mathcal{G}^{ij}(0)+O(|\Delta|)$.

1. **Angular integral.** With $\Delta=r(\cos\phi,\sin\phi)$,

$$\delta^{ij}-2\hat\Delta^i\hat\Delta^j=-\begin{pmatrix}\cos 2\phi & \sin 2\phi\\ \sin 2\phi & -\cos 2\phi\end{pmatrix},\qquad \int_0^{2\pi}d\phi\,e^{-iqr\cos(\phi-\phi_q)}e^{2i\phi}=-2\pi J_2(qr)e^{2i\phi_q},$$

   so

$$\int_0^{2\pi}d\phi\,e^{-i\mathbf{q}\cdot\Delta}\big(\delta^{ij}-2\hat\Delta^i\hat\Delta^j\big)=4\pi\,J_2(qr)\Big(\hat q^i\hat q^j-\frac{1}{2}\delta^{ij}\Big). \tag{3.2}$$

2. **Radial integral.** $\int_0^\infty dr\,r\,J_2(qr)\,\mathcal{G}^{ij}(0)\,\chi(r)=\frac{2}{q^2}\mathcal{G}^{ij}(0)+O(q^{-3})$ for any smooth χ with $\chi(0)=1$, because $\int_0^\infty dt\,t\,J_2(t)=2$. The $O(|\Delta|)$ terms of $\mathcal{G}$ give $O(q^{-3})$.
3. Hence $F=\frac{8\pi}{q^2}\big(\hat q^i\hat q^j-\frac{1}{2}\delta^{ij}\big)\mathcal{G}^{ij}(0)+O(q^{-3})$. With $q=|\mathbf{k}|/\xi$ and $\hat q=\hat{\mathbf{k}}$, this is Eq. (2.1).

- Only the traceless part of $\mathcal{G}^{ij}(0)$ enters. The trace $\mathcal{G}^{ii}(0)$ (which is sensitive to large y) drops out, so $\mathcal{C}$ is finite.
- The $1/q^2$ comes from the discontinuity of $\hat\Delta^i\hat\Delta^j$ at $\Delta=0$. It is a power, not a logarithm.

Contrast: where the $\log\frac{\vee}{k^+}$ of Row IV is. It is in the first line of Eq. (fin), not in this term.

- There the $\frac{1-\xi}{\xi}$ of P_{gg} comes with $\frac{d\xi}{\xi^2}$, and the weight is $-\log\frac{(y'-y)^2\mu^2}{4}$.
- Since $\int d^2\Delta\,e^{-i\mathbf{q}\cdot\Delta}(-\log\Delta^2)\,\mathcal{G}^{ii}(\Delta)=\frac{4\pi}{q^2}\mathcal{G}^{ii}(0)+\dots$, the integrand is $\frac{1-\xi}{\xi^3}\cdot\frac{4\pi\xi^2}{\mathbf{k}^2}\mathcal{G}^{ii}(0)\to\frac{4\pi}{\mathbf{k}^2}\frac{\mathcal{G}^{ii}(0)}{\xi}$.
- This gives $\log\frac{\vee}{k^+}\,\frac{4\pi}{\mathbf{k}^2}\mathcal{G}^{ii}(0)$. Here $y=y'$ $(=z)$ and $\frac{4\pi}{\mathbf{k}^2}$ is the soft emission from z: this is the BFKL coefficient $P_{\text{II.IV}}=2N_c\int K(w,w',z)K(x,x',z)\mathcal{O}(z,z)$ of `LogVeeKplus_BFKL.pdf`.
- The difference: the BFKL piece has the trace $\mathcal{G}^{ii}(0)$ with $\frac{d\xi}{\xi^3}$; the $d_\perp$ term has the traceless part with $\frac{d\xi}{\xi}$.
- In p^+ language (`RowIV_dperp.pdf`, Step 6): for $p^+\gg k^+$ only the $\delta_{mm'}\delta_{kk'}$ part of the polarization sum (c_2) survives. The $d_\perp$ term comes from $c_1+c_3=\frac{d_\perp p^+}{P^2}$, i.e. the weight $\xi(1-\xi)$, which after the z integral falls like $1/p^{+3}$.

4 Remarks on the plus prescription

- Any subtraction is an identity: $\int \frac{d\xi}{\xi} f = \int \frac{d\xi}{\xi}\big[f(\xi) - f(c)\big] + f(c)\log\frac{\xi_1}{\xi_0}$. The only natural choices, $c = 0$ (at the $\vee$ end) or the $p^+ \to 0$ end, give $f(c) = 0$, so nothing changes.
- Do *not* subtract the $\xi = 1$ value at the $\xi \to 0$ end, i.e. $\int_{\xi_0}^{\xi_1} \frac{d\xi}{\xi}\big[f(\xi) - f(1)\big] + f(1)\log\frac{\xi_1}{\xi_0}$. It is exact, but it creates the pair $\pm F(1)\log\frac{\xi_1}{\xi_0} = \pm F(1)\big[\log\frac{\vee}{k^+} - \log(1 + \frac{\Lambda}{k^+})\big]$. The + part would look like an evolution logarithm, but $F(1)$ belongs to neither BFKL nor JIMWLK; it cancels against the − part hidden in the subtracted integral.
- Do not confuse this with $\frac{\xi}{(1-\xi)_+}$ in the first line of Eq. (fin). That one is needed: it regulates the soft end of P_{gg}, and its $\delta(1-\xi)$ term carries $\log\frac{\vee}{\Lambda}$ (JIMWLK). The present term has no $\frac{1}{1-\xi}$.

5 Numerical check

Script `plus_check.py`, results in `plus_check.json`.

Configuration.

- $SU(2)$ in the adjoint ($SO(3)$ rotations); $U(y)$ smooth, $U \to 1$ outside $|y| \approx 3$.
- Two Gaussian colour charges. The colour of the second is chosen so that $\int_x [1 - U(x)]^{ce}\rho^e(x) = 0$; then $\int_x X(x-y)[U(y) - U(x)]\rho(x)$ falls like $1/y^2$.
- 3072^2 grid on a 40×40 box; $\mathcal{G}^{ij}(\Delta)$ by FFT. A smooth window $e^{-\Delta^2/(2\cdot 7^2)}$ only changes F at $q \lesssim 1/7$.
- Normalization: $\mathcal{G}^{ii}(0) = 1$. Here $\mathcal{G}^{ij}(0) = \begin{pmatrix} 0.7273 & -0.1576 \\ -0.1576 & 0.2727 \end{pmatrix}$.

(a) Calibration. For $\mathcal{G}^{ij} = M^{ij}e^{-\Delta^2}$ the grid transform agrees with the exact $4\pi M^{ij}(\hat{q}^i\hat{q}^j - \frac{1}{2}\delta^{ij})\int_0^\infty dr\, r\, J_2(qr)e^{-r^2}$ to $\max q^2|\delta F| = 6.2 \times 10^{-3}$ ($q \le 40$).

(b) Large $q = |\mathbf{k}|/\xi$. q^2F against $8\pi(\hat{q}^i\hat{q}^j - \frac{1}{2}\delta^{ij})\mathcal{G}^{ij}(0)$ ($d_\perp$ term), and q^2F_E (F with $-\log\Delta^2$ and the trace, the BFKL piece) against $4\pi\mathcal{G}^{ii}(0) = 4\pi$:

direction ϕ_q	q	q^2F	prediction	ratio	q^2F_E	$/4\pi$
0.0	10	5.72341	5.71182	1.0020	13.0371	1.0375
0.0	20	5.71116	5.71182	0.9999	12.6283	1.0049
0.0	30	5.71289	5.71182	1.0002	12.5364	0.9976
0.0	45	5.71917	5.71182	1.0013	12.4375	0.9897
0.9	10	−5.16780	−5.15392	1.0027	13.1516	1.0466
0.9	20	−5.15979	−5.15392	1.0011	12.6624	1.0076
0.9	30	−5.15637	−5.15392	1.0005	12.5479	0.9985
0.9	45	−5.15338	−5.15392	0.9999	12.4258	0.9888
2.0	10	−0.68050	−0.73675	0.9237	13.1297	1.0448
2.0	20	−0.72220	−0.73675	0.9803	12.6581	1.0073
2.0	30	−0.73126	−0.73675	0.9926	12.5461	0.9984
2.0	45	−0.73872	−0.73675	1.0027	12.4250	0.9888

The imaginary parts are below 10^{-3} of the real parts.

(c) The lower limit $\xi_0 = k^+/\vee$. $|\mathbf{k}| = 1.5$, $\hat{\mathbf{k}} = \hat{e}_1$. Here $T(\xi_0) = \int_{\xi_0}^1 d\xi\, \frac{1-\xi}{\xi}F$ (this term) and $T_E(\xi_0) = \int_{\xi_0}^1 d\xi\, \frac{1-\xi}{\xi^3}F_E$ (the BFKL piece). δT is the change from the previous row; the predictions follow from Eq. (2.1) and from $F_E \simeq \frac{4\pi\xi^2}{\mathbf{k}^2}\mathcal{G}^{ii}(0)$.

ξ_0	$T(\xi_0)$	δT	predicted $\delta T \propto \xi_0^2$	$T_E(\xi_0)$	δT_E	predicted $\delta T_E \propto \log$
0.2	0.345147	–	–	4.9778	–	–
0.1	0.377444	0.032297	0.032155	8.4101	3.4323	3.3128
0.05	0.386225	0.008781	0.008779	12.0187	3.6086	3.5920
0.025	0.388513	0.002288	0.002287	15.7086	3.6900	3.7316

- T changes by $\propto \xi_0^2$: each halving of ξ_0 adds a quarter of the previous change. It converges to $T(0) = 0.38929$.
- T_E grows by the same amount, $\approx \frac{4\pi}{\mathbf{k}^2}\log 2$, at each halving: this is $\log\frac{\vee}{k^+}$.

(d) The upper limit $\xi_1 = \frac{k^+}{k^+ + \Lambda}$. Strip $\int_{\xi_1}^1 d\xi \frac{1-\xi}{\xi} F$ against $\frac{(1-\xi_1)^2}{2} F(1)$, with $F(1) = 0.46563$:

Λ/k^+	strip	$\frac{(1-\xi_1)^2}{2} F(1)$	ratio
0.3	2.678×10^{-2}	1.240×10^{-2}	2.160
0.1	2.792×10^{-3}	1.924×10^{-3}	1.451
0.03	2.252×10^{-4}	1.975×10^{-4}	1.140
0.01	2.390×10^{-5}	2.282×10^{-5}	1.047

The ratio tends to 1 as $\Lambda/k^+ \to 0$, and the strip falls like $(\Lambda/k^+)^2$: no $\log \Lambda$.